Baseline Model Report

Tessa Adams | CariSurg MedTech Pathways | July 2026

Objective

The objective was to build and evaluate two baseline machine-learning models for predicting Emergency Severity Index (ESI) triage levels using information available at triage. Logistic regression and a decision tree were compared with a stratified dummy classifier to determine whether the trained models learned useful patterns and performed better than random guessing. The models were evaluated based on how accurately they classified patients, with special attention given to whether they correctly identified the most critical ESI Level 1 patients.

Dataset and feature selection

The dataset included 55,121 emergency department visits and 225 variables. It was divided into a training set containing 80% of the data and a testing set containing the remaining 20%. The split was done carefully so that each ESI level appeared in similar proportions in both sets.

The models used all available numeric variables except the ESI target and information recorded after triage that could reveal the patient's outcome. These variables were removed to prevent data leakage, which happens when a model is given information that would not be available at the time the prediction is made. Non-numeric demographic and administrative variables, such as race, ethnicity, gender, insurance status, and arrival information, were also excluded. Age, triage vital signs, and numeric chief-complaint indicators remained available to the model.

Baseline models

Three models were tested. The dummy classifier provided a simple comparison by making predictions based on how common each ESI level was in the dataset. Logistic regression was trained after putting the numeric features on a similar scale. A decision tree with a maximum depth of five was also tested so that it would remain easier to understand and be less likely to fit the training data too closely. The results are as follows:

Results

Model	**Accuracy**	**Macro F1**	**Weighted F1**	**ESI 1 Recall**
Dummy Classifier	0.375	0.204	0.375	0.00
Logistic Regression	0.683	0.508	0.677	0.25
Decision Tree	0.547	0.207	0.449	0.00

Logistic regression achieved the strongest overall performance. It improved accuracy from 0.375 for the dummy baseline to 0.683 and achieved the highest macro F1 and weighted F1 scores, at

0.508 and 0.677 respectively. It was also the only model to correctly identify any ESI Level 1 encounters. The decision tree performed better than the dummy classifier in accuracy and weighted F1, but its macro F1 score was only slightly higher and its ESI Level 1 recall remained 0.00.

Primary metric: Recall for ESI Level 1

Recall for ESI Level 1 was selected as the primary metric because it measures how many truly critical patients the model correctly identifies. Overall accuracy can be misleading in this dataset because ESI 1 cases are extremely rare, so a model could appear accurate while still missing patients who require immediate treatment. Missing an ESI 1 patient could delay life-saving care and create serious patient-safety consequences. Identifying as many true ESI 1 cases as possible is therefore more clinically important than maximising overall accuracy.

Failure-mode reflection

The most concerning failure mode was under-identification of ESI 1 patients. Logistic regression achieved an ESI 1 recall of 0.25, correctly identifying 4 of the 16 true ESI 1 encounters in the test set and missing the remaining 12. The decision tree and dummy baseline both achieved 0.00 recall for this class. In practice, these errors could assign critically ill patients a lower triage priority and delay assessment or treatment. The models are therefore useful as benchmarks but are not suitable for clinical deployment without substantial improvement.

Conclusion

The project established an interpretable baseline for ESI prediction and showed that the selected triage-time features contain useful predictive information. Logistic regression was preferred because it produced the highest accuracy, macro F1, weighted F1, and ESI 1 recall. However, its ESI 1 recall of 0.25 remains inadequate.